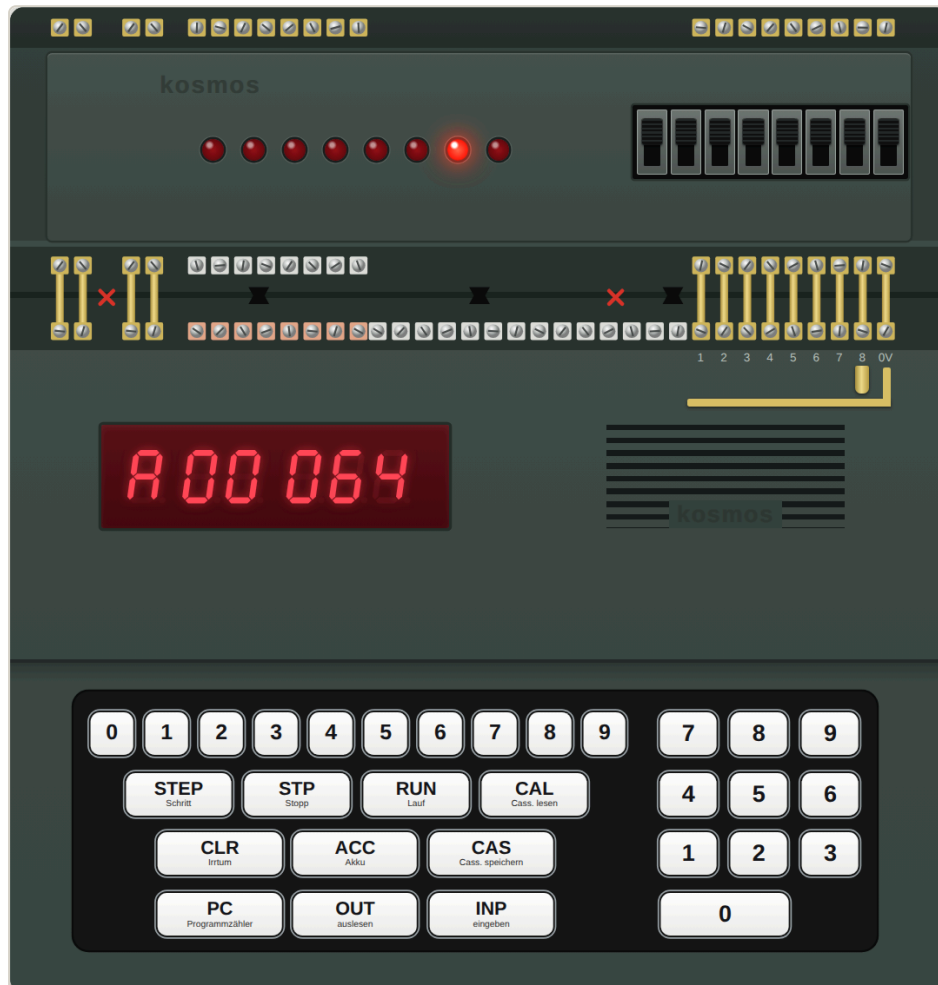


Kosmos CP1 — a programmer's guide

How to operate and program the 1983 Kosmos CP1 experimental computer, written for the [browser emulator](#), which runs the machine's original firmware. Everything described here was tried on that firmware. This is an independent guide in its own words, not a translation of the Kosmos manual; the original German manual is on [archive.org](#).

1. The machine

The CP1 is a small teaching computer with a six-digit display and a calculator-style keypad. You program it in decimal numbers: there is no screen, no letters and no hexadecimal. Under the hood an Intel 8049 microcontroller runs a fixed program (the firmware) that makes the box behave like a much simpler imaginary computer. That simple computer is what you program, and it has just four parts worth knowing.



The CP1 with its input/output module, as drawn in the emulator. Display at the left, keypad below.

- **Memory.** 256 numbered cells, 000 to 255 (128 in the basic machine, 256 with the memory expansion, which the emulator has fitted). A cell holds five digits, written as two plus three: 04 . 200. Program and data share the same cells.
- **The Akku** (accumulator). The one working register. Almost every instruction either puts something into the Akku, takes something out of it, or changes it.
- **The program counter.** The number of the cell that will be carried out next.
- **Two ports.** Port 1 and Port 2, eight lines each, connect the computer to the outside world.

Instructions and numbers look the same

A cell's five digits are read as *instruction code* (two digits) and *operand* (three digits). 04.200 means instruction 04 with operand 200. A plain number is simply a cell whose first two digits are 00: 00.017 is the number 17. Numbers range from 0 to 255. There are no negative numbers and no fractions; a calculation that leaves this range stops the program with an error.

2. Operating it

When switched on, the display shows **P 000**: the program counter is at cell 000. The letter at the left always tells you what you are looking at.

Display	Meaning
P 012	the program counter: the next cell to run is 012
E04042	entry : this is what you just stored (or, as E 012 , where the next entry will go)
C04042	the contents of a cell you asked to see
A00042	the Akku
F 006	an error (<i>Fehler</i>); see section 8
123	digits you are typing, not yet used

The keys

Keys	What happens
three digits, INP	Sets where entries will be stored. 0 1 2 INP shows E 012 .
five digits, INP	Stores the five digits in that cell and moves on to the next cell, so a program is typed as one line after another.
three digits, OUT	Shows the contents of that cell.
OUT alone	Shows the next cell; press it repeatedly to read a program back.
three digits, PC	Sets the program counter: where RUN will start.
PC alone	Shows the program counter.
RUN	Runs the program from the program counter.
STP	Stops a running program. RUN continues where it stopped.
STEP	Carries out one instruction and stops; the display shows the new program counter.
ACC	Shows the Akku.
CLR	Clears what you have typed, or an error display.
CAS , CAL	Save to and load from cassette. Not available in the emulator; use its Save and Open buttons.

Entering a program therefore goes: address, **INP**, then five digits and **INP** for each line. Starting it goes: address, **PC**, **RUN**. Note that these are two separate pointers: **INP** says where you write, **PC** says where the computer runs. Typing the wrong number of digits before a key gives **F 001**.

In the emulator you can skip the typing: pick a program and press **Load & run**, or click a line in the **Inside** panel and edit the cell directly. Switching the machine off and on (the **Esc** key) clears all cells, as it does on the real one.

3. The instructions

There are 21 instructions on the basic machine, plus three for expansion ports. Each has a two-digit code and a three-letter name; the names are abbreviations of German words, given in the last column. "Cell *n*" means the cell whose number is the operand.

Code	Name	What it does	From
01.000	HLT	Stop. The display shows the program counter.	Halt
02.000	ANZ	Show the Akku on the display.	Anzeigen (display)
03.nnn	VZG	Wait n milliseconds (1–255).	Verzögern (delay)
04.nnn	AKO	Put the number n itself into the Akku.	Akku, Konstante (constant)
05.nnn	LDA	Copy cell n into the Akku.	Laden (load)
06.nnn	ABS	Copy the Akku into cell n .	Abspeichern (store)
07.nnn	ADD	Add cell n to the Akku.	Addieren
08.nnn	SUB	Subtract cell n from the Akku.	Subtrahieren
09.nnn	SPU	Jump: continue at cell n .	Sprung, unbedingt (unconditional)
10.nnn	VGL	Compare: is the Akku equal to cell n ?	Vergleich, gleich
11.nnn	SPB	Jump to cell n if the last comparison was true; otherwise carry on.	Sprung, bedingt (conditional)
12.nnn	VGR	Compare: is the Akku greater than cell n ?	Vergleich, größer
13.nnn	VKL	Compare: is the Akku less than cell n ?	Vergleich, kleiner
14.000	NEG	Turn a 0 in the Akku into 1 and a 1 into 0.	Negieren
15.nnn	UND	Logical AND: the Akku becomes 1 only if both it and cell n are 1.	Und (and)
16.00n	P1E	Read Port 1 into the Akku. $n = 0$: all eight lines as one number; 1–8: that line alone, as 0 or 1.	Port 1, Eingabe (input)
17.00n	P1A	Write the Akku to Port 1. $n = 0$: all lines; 1–8: that line alone.	Port 1, Ausgabe (output)
18.00n	P2A	Write the Akku to Port 2, in the same two ways.	Port 2, Ausgabe
19.nnn	LIA	Load indirectly: cell n holds an address; copy the cell at <i>that</i> address into the Akku.	Laden, indirekt
20.nnn	AIS	Store indirectly: copy the Akku into the cell whose address is in cell n .	Akku indirekt speichern
21.nnn	SIU	Jump indirectly: continue at the address held in cell n .	Sprung, indirekt
22–24	P3E, P4A, P5A	Read Port 3, write Port 4 and Port 5 of an expansion module. In the emulator P4A drives the tone generator (section 7).	

Things worth knowing

- **AKO versus LDA.** 04.007 puts the number 7 into the Akku. 05.007 puts *whatever is in cell 007* into the Akku. Mixing them up is the classic first mistake.
- **Arithmetic is on plain numbers only.** ADD and SUB need a plain number (a cell starting with 00) on both sides, and the result must lie between 0 and 255. Adding 100 to 200 or taking 9 from 5 stops the program with **F 006**.
- **Comparing does not change the Akku.** VGL, VGR and VKL only remember "true" or "false" for a following SPB. An SPB without a comparison before it does not jump.
- **LDA and ABS move the whole cell**, all five digits, so a program can copy instructions as well as numbers.
- **NEG and UND work on 0 and 1 only.** Anything else in the Akku gives **F 005**.
- **After HLT the display shows the program counter**, not your result. Press **ACC** to see the Akku, or end the program with a jump to itself so the last ANZ stays on the display.

4. First programs

Each program below is in the emulator's library under "Examples from the guide". The lines are written as the address, the name of the instruction, and the five digits you would type.

Show a number

```
000 AK0 04.042   Akku := 42
001 ANZ 02.000   show the Akku
002 SPU 09.002   jump to this same line: stay here
```

Run it and the display shows **A00042**. The last line is a loop on the spot; **STP** gets you out. [Try it](#)

Add two numbers

```
000 LDA 05.010   Akku := cell 010
001 ADD 07.011   Akku := Akku + cell 011
002 ABS 06.012   keep the sum in cell 012
003 ANZ 02.000
004 SPU 09.004
010     00.017   the first number
011     00.025   the second number
```

Data lives in cells of its own, here 010 to 012, well away from the program. The display shows 42. Change cell 010 or 011 and run it again. [Try it](#)

A loop with an end: count to ten

```
000 AK0 04.000
001 ABS 06.020   counter := 0
002 LDA 05.020
003 ANZ 02.000   show the counter
004 VZG 03.250   wait a quarter of a second
005 VGL 10.021   is the counter equal to the limit?
006 SPB 11.010   yes: leave the loop
007 ADD 07.022   no: counter := counter + 1
008 ABS 06.020
009 SPU 09.002   round again
010 SPU 09.010   finished
021     00.010   the limit
022     00.001   the step
```

This is the pattern for every loop on the CP1: do the work, compare, jump out if done, otherwise change the counter and jump back. There is no "add 1" instruction, so the 1 has to sit in a cell. Without the VZG the count

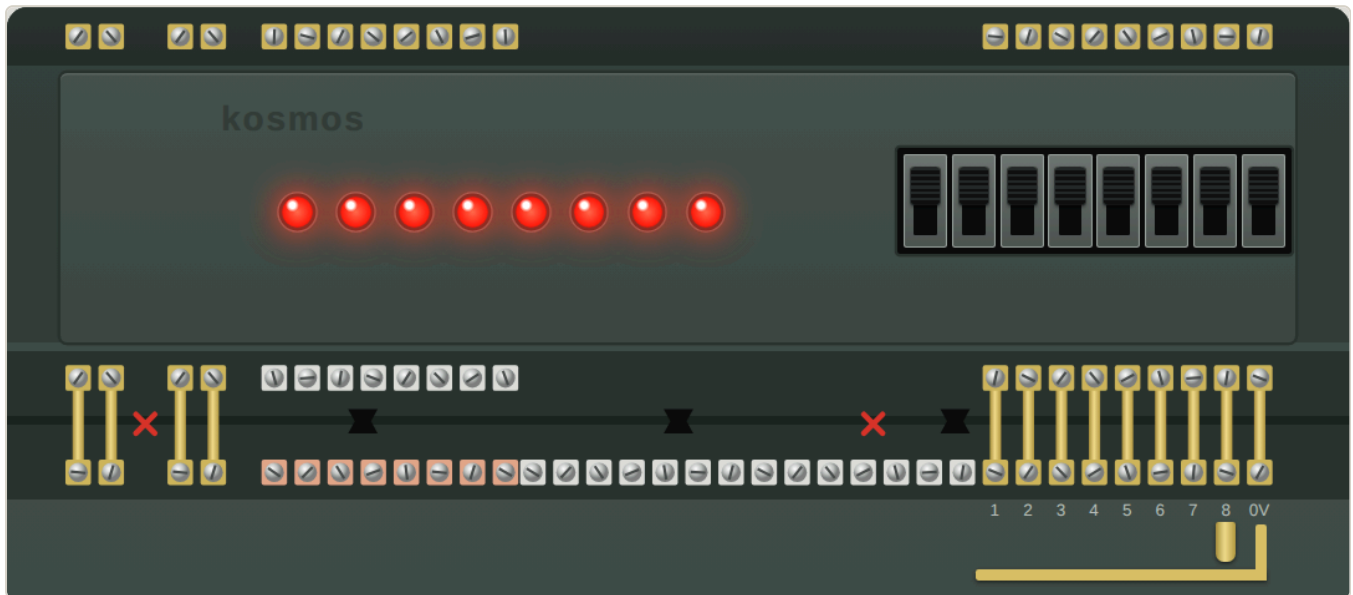
would flash by too fast to read. [Try it](#)

Multiplying

The CP1 can only add and subtract. To multiply 12 by 9 you add 12 nine times, counting down as you go. The library program "Multiplying by repeated addition" does exactly that and shows the running total: 12, 24, 36 ... 108. Division works the other way round, by subtracting and counting how often it fits. [Try it](#)

5. Ports: inputs and outputs

The ports are rows of screw terminals along the top edge of the computer. Each port has eight lines, numbered 1 to 8. A line is either at 5 volts, which the computer calls 1, or at 0 volts, which it calls 0.



The input/output module above the computer: eight LEDs on Port 2, eight switches on Port 1.

Eight lines make one number

Read or written all together, the eight lines form a binary number: line 1 counts 1, line 2 counts 2, line 3 counts 4, then 8, 16, 32, 64 and line 8 counts 128. All eight lines at 1 make 255. That is why 255 is the largest number the CP1 knows.

Port 1: inputs

16.000 (P1E) reads all eight lines into the Akku as one number. 16.003 reads line 3 alone and gives 0 or 1. A line with nothing connected reads 1. To make it 0 you connect it to the 0 V terminal. The emulator offers two ways:

- **The switches** on the module. Up is 1, down is 0, and a switch stays where you put it.
- **Contact clips.** A springy brass clip screwed to a terminal, hanging over a rail connected to 0 V, makes a push button: press it and the line is 0, let go and it is 1 again. In the Ports panel you choose which lines have a clip; click a clip, or hold **Shift** and the line's number.

Port 1 can also be written, with 17.00n (P1A). The emulator can show small lamps on its terminals for programs that do so.

Port 2: outputs

18.000 (P2A) puts the Akku on all eight lines; the LEDs show it in binary. 18.003 sets line 3 alone from an Akku of 0 or 1. After switching on, all Port 2 lines are at 1, so all LEDs are lit.

Echo: switches to LEDs

```
000 P1E 16.000   Akku := the switches
001 P2A 18.000   LEDs := Akku
002 ANZ 02.000   and show the number
003 SPU 09.000
```

Flip switches and watch the LEDs and the number follow. [Try it](#)

A running light

One lit LED walking along Port 2 means the values 1, 2, 4, 8 ... 128. Doubling is adding a number to itself, so no multiplication is needed: load the value, add the same cell again, store it. At 128 the program starts over. [Try it](#)

A push button

```
004 P1E 16.008   Akku := line 8 (1 = open, 0 = pressed)
005 VGL 10.021   still 1?           (cell 021 holds 1)
006 SPB 11.004   yes: keep waiting
```

That loop waits for a press. The library program "Counting button presses" then adds one to a counter and waits in a second loop until the button is released again. Without that second wait, one press would be counted hundreds of times. [Try it](#)

Scrambled numbers

The CP1 has no random-number instruction. A well-known trick is to wire the eight Port 2 lines to the eight Port 1 lines in a mixed-up order. A program then writes a counter to Port 2 and reads it straight back from Port 1 with its bits shuffled: 1, 2, 3, 4 come back as 4, 8, 12, 2. The emulator has this as a checkbox, "Wire Port 2 to Port 1", with the order of the wires editable. [Try it](#)

6. Indirect addressing

Normally an instruction names the cell it works on. LIA, AIS and SIU instead name a cell that *contains* the number of the cell to use. That one step of indirection is what makes tables, stacks and subroutines possible, because a program can calculate an address.

Walking through a table

```
004 LIA 19.030   Akku := the cell that cell 030 points at
005 ADD 07.031
006 ABS 06.031   sum := sum + that number
007 LDA 05.030
008 ADD 07.032   (cell 032 holds 1)
009 ABS 06.030   pointer := pointer + 1
```

Cell 030 is the pointer. It starts at 040, the first table entry, and is moved on by one each round, so the same three instructions visit 040, 041, 042 and so on. The full program adds up five numbers and shows 75. [Try it](#)

Subroutines

There is no call instruction. The convention is: before jumping to a subroutine, put the address you want to return to in an agreed cell; the subroutine ends with SIU on that cell.

```
000 AK0 04.003   the address to come back to
001 ABS 06.050   ... goes into cell 050
002 SPU 09.030   jump to the subroutine
003 ...          execution continues here afterwards

030 ...          the subroutine
036 SIU 21.050   return: jump to the address in cell 050
```

Note the AK0: the return address is a constant, the number 3, not the contents of cell 003. A subroutine that calls another one needs a second return cell, or a stack built with AIS and LIA, which is how the recursive Towers of Hanoi in the library works. [Try it](#)

7. Sound

Programs make sound by switching tone generators connected to a port. The emulator provides two arrangements.

- **A tone generator on Port 4.** 23.000 (P4A) writes the Akku to Port 4. 0 is silence; in the emulator 1 is middle C and each step up is a semitone higher, so 1 to 13 is one octave. This pitch mapping is the emulator's own choice. Programs: "Sound: scale" and "Sound: melody". [Try the scale](#)
- **One tone per Port 2 line.** With the option "Tone generators on Port 2", line 1 sounds c, line 2 d, then e, f, g, a, b and line 8 the c above. Several lines at once give a chord. "Sound: switch piano" turns the eight Port 1 switches into keys this way. [Try the piano](#)

A melody is just a table of notes played through with LIA, a VZG for the length of each note, and a silence between notes so that two equal notes can be told apart. The **Sound off** button silences whatever is sounding.

8. Error codes

When something goes wrong the computer stops and shows **F 00n**. **CLR** clears the display; **PC** shows where it stopped.

Code	Cause	Typical case
F 001	Wrong number of digits for the key pressed.	Two digits and PC ; six digits; one digit and RUN .
F 002	The cell to be run is not an instruction.	Running into an empty cell or into data, often after a wrong jump or a missing HLT.
F 003	An address beyond the memory fitted.	Cell 200 on a machine without the memory expansion.
F 004	Something that cannot be stored or addressed.	An instruction code above 24, an operand above 255, an address above 255.
F 005	A value that does not fit the instruction.	Arithmetic on a cell that holds an instruction; NEG or UND on something other than 0 or 1; a port line number above 8.
F 006	A result outside 0–255.	ADD going above 255, SUB going below 0.
F 007	Cassette loading failed.	Does not occur in the emulator.

9. Built-in demos

Two small programs are part of the firmware and need nothing entered:

- **9** **RUN**: a countdown. The display runs from 999999 through 888888 down to 000000.
- **8** **RUN**: a reaction test. After a pause a number races up from 0 towards 255, about one count per millisecond; **STP** freezes it. The pause gets longer each time you run it.

10. About the emulator

The [emulator](#) runs the CP1's original 8049 firmware, with the two 8155 memory and port chips, the display and the keypad modelled around it. What it adds for convenience:

- **Programs:** a library, loading and saving program files, and **Load & run**. Loading a program also sets up the wiring it needs.
- **Inside:** the program counter, the Akku, the ports line by line, and all 256 cells with their instruction names; click a cell to change it.
- **Speed:** slower or faster than the original.
- **PC keyboard:** digits, **Enter** or **I** for INP, **O** OUT, **P** PC, **A** ACC, **R** RUN, **.** or **H** STP, **T** STEP, **X** CLR, **Esc** power off and on.

Program files are plain text, one cell per line: 012 ako 04.200 or just 012 04.200; text after # is a comment.

Not emulated: the cassette interface, Ports 3 and 5, and add-on circuits other than those described above.

Quick reference

Code	Name	Effect	Code	Name	Effect
01	HLT	stop	12	VGR	Akku > cell?
02	ANZ	show Akku	13	VKL	Akku < cell?
03	VZG n	wait n ms	14	NEG	0 ↔ 1
04	AKO n	Akku := n	15	UND	Akku AND cell
05	LDA	Akku := cell	16	P1E n	Akku := Port 1 (line n)
06	ABS	cell := Akku	17	P1A n	Port 1 (line n) := Akku
07	ADD	Akku + cell	18	P2A n	Port 2 (line n) := Akku
08	SUB	Akku – cell	19	LIA	Akku := cell pointed at
09	SPU	jump	20	AIS	cell pointed at := Akku
10	VGL	Akku = cell?	21	SIU	jump to address in cell
11	SPB	jump if true	23	P4A	Port 4 := Akku (tone)

Keys: address **INP** sets the entry position · five digits **INP** stores · address **OUT** shows a cell · address **PC** **RUN** starts · **STP** stops · **STEP** single step · **ACC** shows the Akku · **CLR** clears.

Written with Claude Opus 5.5 (Claude Code) for the [Kosmos CP1 emulator](#) by Michael Wessel.
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